

ADVANCED CALCULATOR

Professional Multi-Functional Calculator Application

Next Hikes IT Solutions

Presented by

Aditi Umale



Introduction

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Simple for everyday use, powerful when more is needed.

What is Advanced Calculator?

Advanced Calculator is a desktop application built with Python and Tkinter. It brings everyday arithmetic and advanced calculation tools together in one clean interface.

Why it is useful

- Quick daily arithmetic and percentage calculations
- Scientific functions for students and technical users
- Financial, BMI and conversion utilities
- Calculation history for reviewing previous results
- Light/dark mode and easy three-line navigation

Daily-life examples

- 01 Bills & totals
- 02 Study & homework
- 03 Quick percentages
- 04 BMI & health
- 05 Conversions
- 06 Financial checks

Tools, Libraries & Links

03

Development stack and project resources.

Development stack

Python 3	Programming language
Tkinter / ttk	Desktop GUI framework
Visual Studio Code	Development environment
Git & GitHub	Version control and sharing

Useful resources

- ↗ **Python Documentation**
<https://docs.python.org/3/>
- ↗ **Tkinter Documentation**
<https://docs.python.org/3/library/tkinter.html>
- ↗ **Visual Studio Code**
<https://code.visualstudio.com/>
- ↗ **GitHub**
<https://github.com/>
- ↗ **Chatgpt**
<https://chatgpt.com/c/6a8a986f-a1e4-83ee-89f4-8b1b246d5539>

Challenges & Solutions

04

Turning real development problems into working features.

01 Scientific expressions

Challenge:

Reliable handling of sin, cos, tan and logarithms.

Solution:

Dedicated scientific logic with degrees by default.

02 Invalid calculations

Challenge:

Incomplete expressions could trigger an error.

Solution:

Improved validation and scientific function behavior.

03 Calculation history

Challenge:

Users needed to review previous results.

Solution:

Added History with clear and export options.

04 Clean navigation

Challenge:

Many tools can make the UI confusing.

Solution:

Grouped tools in a three-line menu.

Learning & Exploration

05

What the project taught me beyond writing the code.

What I learned

- Python GUI development with Tkinter and ttk
- Event-driven programming and button interactions
- Mathematical expression handling and error handling
- History management and file export
- Light/dark theme and professional UI design
- Debugging, testing and iterative improvement
- GitHub-based project organization

Where I explored

- Python and Tkinter documentation
- Visual Studio Code development tools
- Git and GitHub for version control
- Python mathematical libraries and functions
- AI-assisted debugging and development
- User-interface design and testing

Key takeaway

Programming is not only about writing code — it is also about testing, debugging, solving problems and improving the user experience.

THANK YOU

Advanced Calculator

Simple calculations. Advanced possibilities.

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